

Jordan Thompson

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WORK EXPERIENCE

VRNDA Labs

June 2026 - Present

Junior Mechanical Engineer

Richmond, BC

- Engineered fluid delivery systems utilizing NEMA 17 stepper motors, CL42T-V41 drivers, and 3D printed parts designed in Fusion360; achieving 0.1–1200 mL/hr flow rates via microstepping at 51,200 steps/rev.
- Designed a minimum viable prototype for a modular syringe pump; from requirements to a CAD design with a bill of materials in 2 weeks.
- Produced Fusion360 assembly drawings for an end effector hot-swap mechanism, applying GD&T, dimensioning, and tolerance analysis with a focus on design for manufacturing and assembly.

Kardium Inc

May 2025 – Aug 2025

Biomedical Test Engineer Co-op

Burnaby, BC

- Designed and executed design verification test plans to validate cleaning and sterilization efficacy for custom manufactured and vendor-sourced parts to generate data that directly supported a FDA regulatory submission.
- Ensured ISO 10993 biocompatibility compliance within an ISO 13485 Quality System; conducted non-parametric statistical analysis in Minitab and participated in change controls, ECOs, and deviations.
- Reduced TOC test cost by 70% by leading the purchase of an automated TOC analyzer. Collaborated with outside vendors, and ran test protocols to complete the IQ/OQ/PQ process under GMP guidelines.

Bioaugmentative Interfaces Lab - [Co-Authored Publication](#)

Nov 2023 - April 2025

Mechatronics Co-op Student

Vancouver, BC

- Designed and built a phantom bladder rig (3D-printed stand, flow meter, and force sensing) to mechanically validate flow rates and actuation performance of an implantable electromechanical device ahead of in-vivo animal trials.
- Contributed to the mechanical design and packaging of a device housing, supporting migration to a miniaturized form factor for rodent implantation.

Mechatronics Research Assistant

Vancouver, BC

- Optimized a shape-memory alloy actuator's duty cycle, achieving 78% liquid voiding efficiency while restricting temperature increases to <2°C.
- Designed modular 3D-printed molds in OnShape to improve manufacturing repeatability, reduce air bubbles, and streamline actuator insulation. Cut build time by 25% and reduced overall defect rate.

SKILLS

CAD & Prototyping: SolidWorks, Fusion360, OnShape, GD&T, 3D printing, Laser cutting, Hand tools

Quality & Regulatory: ISO 13485, ISO 10993, 21 CFR, GMP, IQ/OQ/PQ, DHF, Minitab

Programming: C/C++, MATLAB, Python

Electronics & Test: Altium, Eagle, Oscilloscope, DMM, Soldering, Data Acquisition

PROJECTS

Dr. Wobbles, Open Source Pediatric Pain Relief Device - [Website](#)

2025 - 2026

- Designed a screwless, modular 3D-printed housing in OnShape, enabling non-technical user assembly in under 13 minutes for an open-source medical device that provides pain relief through vibration and cooling.
- Engineered a motor-driving PCB that delivers stable varying vibrational frequency modes (68–172 Hz) with a brushed DC motor, using noise filtering techniques to prevent MCU brownouts (lowpass filter + flyback diode).
- Authored a Design History File covering design requirements, test protocols, and drawings, aligned with medical device documentation practices.

Spatio-Temporal Electromyography Data Analysis Algorithm

2025 - 2026

- Cut data processing time from 3 months to minutes with an automated MATLAB detection algorithm within 1% of human labelling accuracy.

EDUCATION

University of British Columbia BSc, Biomedical Engineering specialized in Mechatronics, Graduated April 2026